

Scripting using PHP (Hyper Text Pre Processor)

DAY (1 to 8)	Topic	Actual Date	Planned Date	Status (Completed)	Comment
1	Introduction to World Wide Web				
	What is URL, Client, Server ,Web pages (Static vs Dynamic)				
	Website vs Web application				
	Introduction of Procedural Oriented Programming				
	Variables				
	Variable Initialization				
	Print & Echo				
	Functions				
	Conditional Statements Loops (For, While, For each)				
	Arrays (Indexed, Associative)				
	Creating Forms (Post & Get) Processing Form Data (Post & Get)				
2	Practical				
3	Database Connection				
	Executing Queries				
	Registration Module				
	Form Validation				
4	Practical				
5	User Management Module				
	Login Module				
	Session (Login & Logout) Features				
6	Practical				
7	Object Oriented Programming (APIE , Static , Abstract and Interface)				
8	File Handling (Add , Show)				
9	Practical				
10	Website Integration (Combine HTML, CSS, Bootstrap and PHP to create dynamic web pages)				
11	Excel Downloader				
12	Pdf Downloader				

I acknowledge that I am fully satisfied with the session and have gained a clear understanding of all the topics covered.

Sign: _____

Classwork 1

No.	Practical Questions	Status (Completed)
1	Write a PHP script that uses a foreach loop to iterate through an associative array of 5 different fruits and their colors.	
2	Write a function named calculateArea() that takes two parameters: length and width. The function should return the area of a rectangle (length * width).	
3	Create an HTML form with the following fields: - A text input for the user's name. - A number input for the user's age. - A dropdown select for the user's gender (Male, Female, Other). - A submit button. - Use the POST method to send the form data to a PHP script. In the PHP script, process the form data and display a message like: O/P: Hello, [Name]. You are [Age] years old and identify as [Gender].	
4	Create an HTML form with the following fields: - A text input for the user's email. - A password input for the user's password. - A checkbox for subscribing to a newsletter. - Use the GET method to send the form data to a PHP script. In the PHP script, process the form data and display a message based on whether the user subscribed to the newsletter. For example: O/P: Thank you for signing up, [Email]. You have [subscribed/not subscribed] to the newsletter.	
5	Write a PHP script named array_looping.php that: - Use a for loop to iterate over an index-based array and display its elements. - Use the foreach() loop to iterate over an associative array and display its elements. - Looping with for and foreach(). - Demonstrate the use of array library functions such as array_push(), array_pop(), array_merge(), array_slice(), and array_keys().	
6	Write a PHP script named strings.php that: - Demonstrate how to create and access strings using single quotes and double quotes. - Show how to concatenate strings and display the result. - Demonstrate string searching using strpos() and displays the position of a substring. - Demonstrate string replacing using str_replace() and displays the modified string. - Demonstrate the use of string library functions such as strlen(), strtolower(), strtoupper(), trim(), and substr().	

Homework 1

No.	Practical Questions	Status (Completed)
1	Create Write a PHP script that uses a foreach loop to iterate through an associative array of 5 different car models and their brands. Display each car model and its brand.	
2	Write a function named CalculatePerimeter() that takes two parameters: length and width. The function should return the perimeter of a rectangle (2 * (length + width)).	
3	Create an HTML form with the following fields: - A text input for the user's full name. - A text input for the user's phone number. - A dropdown select for the user's preferred car brand (Toyota, Ford, Tesla). - A submit button. Use the POST method to send the form data to a PHP script. In the PHP script, process the form data and display a message like: O/P: Hello, [Full Name]. Your phone number is [Phone Number] and your preferred car brand is [Car Brand].	
4	Create an HTML form with the following fields: - A text input for the user's username. - A password input for the user's password. - A checkbox for agreeing to the terms and conditions. Use the GET method to send the form data to a PHP script. In the PHP script, process the form data and display a message indicating whether the user agreed to the terms. For example: O/P: Welcome, [Username]. You have [agreed/not agreed] to the terms and conditions.	
5	Write a PHP script named car_array_loop.php that: - Use a for loop to iterate over an index-based array of car models and display its elements. - Use the foreach() loop to iterate over an associative array of car models and their prices, displaying both. - Demonstrate the use of array library functions like array_push(), array_pop(), array_merge(), array_slice(), and array_keys() to manipulate the arrays.	
6	Write a PHP script named car_strings.php that: - Demonstrates how to create and access strings with single quotes and double quotes for car descriptions. - Shows how to concatenate strings like car models and their prices, displaying the result. - Demonstrates searching for a substring within a car description using strpos() and displays the position of the substring. - Demonstrates replacing a substring in a car description using str_replace() and displays the modified string. - Demonstrates the use of string library functions such as strlen(), strtolower(), strtoupper(), trim(), and substr() to manipulate car descriptions.	

Classwork 2

No.	Practical Questions	Status (Completed)
1	Create a PHP script that establishes a connection to a MySQL database using the mysqli. It should include host, username, password, database_name.	
2	Perform following CRUD operation in php. - Insert Data in Db. - Update Data in Db. - Delete Data in Db. - Show Data in Db.	
3	Write a PHP script to establish a connection to a MySQL database using mysqli. - Implement the following CRUD operations for an "employees" table: - Insert Data: Add a new employee with fields like ID, name, department, and salary. - Update Data: Modify the department of an employee. - Delete Data: Remove an employee record. - Show Data: List all employees in an HTML table format.	
4	Create a PHP-based blog posting form that includes the following fields: - Title (text input), Content (textarea), Author (text input), Publish Date (date input) - Validate the form data (e.g., fields are not empty, date format is correct). - Store the blog post data in a MySQL database. - Display a success message upon successful submission, or show an error message if any validation fails.	

Homework 2

DAY	Practical Questions	Status (Completed)
1	Create a PHP script that establishes a connection to a MySQL database using mysqli. The script should include host, username, password, and employee_db as the database name	
2	Write a PHP script that performs the following CRUD operations for managing employee data: - Insert Employee Data (name, job title, and salary) into the database. - Update Employee Data (modify the salary of a specific employee by their ID). - Delete an Employee Record by their ID. - Show All Employee Data in a list from the database	
3	Write a PHP script that connects to a MySQL database and performs the following tasks: Create a table named employees with columns: id (primary key, auto-increment), name (VARCHAR), position (VARCHAR), salary (FLOAT). - Insert three records into the employees table (employee names, positions, and salaries). - Retrieve all employee records and display them in an HTML table format.	
4	Create a PHP-based registration form for adding employees to the system. The form should include the following fields: Full Name (text Input), Email (text Input), Password (password Input), Confirm Password (password Input), Job Title (text Input). Validate the form data: - Check if fields are not empty , - Ensure passwords match, - Validate the email format , - Store the employee registration data (name, email, hashed password, and job title) in the MySQL database. Display a success message if registration is successful, or an error message if there are issues (e.g., email already exists).	

CLASSWORK 3

DAY	Practical Questions	Status (Completed)
1	Create a PHP-based login form that includes the following fields: Username (text input), Password (password input): Start a session when the user logs in successfully and store the user's username in a session variable. Create a "welcome" page that displays a personalized greeting using the username stored in the session. Provide a "logout" link on the "welcome" page that destroys the session and redirects the user back to the login page.	
2	Create the registration.html. - Form should be created in bootstrap. - There should be Form with the 4 fields(Name, Email, Username, Password). - When the user submit form action should be registration.php - User should be redirected to Login.php once successfully logged in redirect to home page and generate a dashboard.	
3	Create a PHP-based User Management System that allows for basic CRUD (Create, Read, Update, Delete) operations. - The system should include: A MySQL database with a user table. The table should have columns for id (primary key, auto-increment), username (VARCHAR), email (VARCHAR), password (VARCHAR), and role (VARCHAR). - Create: Provide a form to add new users with fields for username, email, password, and role. - Read: Display a list of all users in a table format, including their username, email, and role. - Update: Allow editing a user's details (except password) through a form. - Delete: Provide a way to delete a user from the system.	
4	Create a PHP script named session_functionality.php that: - Start a session using session_start(). - Set a session variable and displays its value on the page. - Demonstrates retrieving the session variable from another script. - Generate session logout.	

HOMEWORK 3

DAY	Practical Questions	Status (Completed)
1	Create a PHP-based login form for a Project Management System that includes the following fields: Project Manager Username (text input), Password (password input). - When the user logs in successfully, start a session and store the project manager's username in a session variable. - Create a "dashboard" page that displays a personalized greeting using the project manager's username stored in the session. - Provide a "logout" link on the "dashboard" page that destroys the session and redirects the user back to the login page.	
2	Create a registration form in Bootstrap for project managers to signup. The form should include: - Full Name (text input), Email (text input), Username (text input) and Password (password input). - When the form is submitted, It should post the data to project_registration.php. - Upon successful registration, the user should be redirected to project_login.php. - Once logged in, they should be redirected to the dashboard, which lists current projects and tasks.	

3	Create a PHP-based Project Management System that allows for basic CRUD (Create, Read, Update, Delete) operations for managing projects and tasks. The system should include: The table should include columns: id (primary key, auto-increment), project_name (VARCHAR), project_description (TEXT), status (VARCHAR: "pending", "in-progress", "completed"), start_date (DATE), end_date (DATE). • Create: Provide a form to add new projects with fields for project name, Description, status, start date, and end date. • Read: Display a list of all projects in a table format, showing the project Name, status, start and end dates. • Update: Allow editing project details (name, description, status, start date, And end date) through a form. • Delete: Provide an option to delete a project from the system.	
4	Create a PHP script named task_session_functionality.php that: a. Starts a session using session_start(). b. Sets a session variable named task_id and displays its value on the page, representing the currently active task being worked on by a team member. c. Demonstrates retrieving the task_id session variable from another script to track which task is active across different pages	

CLASSWORK 4

No	Practical Questions	Status (Completed)
1	File Handling Operations 1. Read: Write a PHP script to read a file containing a list of employees and their salaries. 2. Write: Add a new employee's name and salary to the file. 3. Append: Add multiple employee records to the file without overwriting the existing content. 4. Delete: Delete the file if all employee data is processed or if it exceeds a specific size.	
2	Student Form with Redirection Create a PHP script named employee_form_redirect.php that: Contains an HTML form with fields for employee name, department, and salary. Redirects the user to a "confirmation" page or another PHP script after the form is submitted. The confirmation page should display a message like: "Thank you, [Name]. Your data has been successfully recorded."	
3	Define a class Employee with the following attributes: 1. id (employee ID) 2. name (employee name) 3. salary (employee salary) 4. Add a method displayDetails() that prints the details of the employee. Create an object of the class and initialize it with specific values (e.g., employee ID, name, and salary).	
5	Create An abstract method calculateBonus() that will be implemented by its subclasses to calculate bonuses based on different criteria (e.g., performance, years of service). Create two classes: 1. FullTimeEmployee: Calculates the bonus based on performance and years of service. 2. PartTimeEmployee: Calculates the bonus based only on performance.	
6.	Define an interface TrainingActions with methods like startTraining(), completeTraining(), and cancelTraining(). Implement this interface in two classes: 1. OnlineTraining: Implements logic specific to online training sessions. 2. InPersonTraining: Implements logic specific to in-person training sessions.	

HOMework 4

No	Practical Questions	Student Name
1	Perform the following operations for file handling (Student Records): - Read: Read a file that contains a list of student names and grades. - Write: Write a new student's name and grade to the file. - Append: Append additional student records to the file without overwriting the existing data. - Delete: Delete the file that contains student records after processing or upon specific conditions.	
2	Create a PHP script named student_form_redirect.php that: - Contains an HTML form with fields for student name, age, and course. - Redirects the user to a "thank you" page or another PHP script after the form is successfully submitted. The "thank you" page should display a confirmation message such as "Thank you for registering."	
3	Define a class Student with the following attributes: - id (student ID), name (student name), grade (student's final grade). - Add a method displayDetails() that prints out the details of the student. Create an object of the class Student and initialize it with specific values (e.g., student ID, name, and grade).	
4	Create an abstract class StudentRecord for a student record management system. The class should have: An abstract method calculateGrade() that will be implemented by its subclasses to calculate grades based on different criteria (e.g., assignments, exams). Create two classes: - UndergraduateStudent that calculates the grade based on assignments and exams. - GraduateStudent that calculates the grade based on research projects and exams.	
5	Create an interface CourseActions that defines methods like enroll(), drop(), and completeCourse(). Implement this interface in two classes: - OnlineCourse: Implements enroll(), drop(), and completeCourse() with specific logic for online courses. - InPersonCourse: Implements enroll(), drop(), and completeCourse() for in-person courses.	